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PI Name:	Mudawar, Issam Ph.D.		
Project Title:	Flow Boiling and Condensation Experiment (FBCE)		
Division Name:	Physical Sciences		
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Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
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Zip Code:	47907-1288	**Congressional District:**	4
Comments:			
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Task Description:	Several efforts are presently underway to reduce the size and weight of space systems by transitioning from single-phase to two-phase thermal management. Because of large density differences between liquid and vapor, buoyancy can play an important role in defining the motion of vapor relative to liquid and, therefore, influence heat transfer effectiveness. This is particularly the case for future space missions, including the manned mission to Mars. The emphasis in two-phase fluid physics is evident from several recent NASA workshops that culminated in critical recommendations concerning Rankine cycle power conversion, thermal control systems, and advanced life support systems. Among others, key fluid physics mechanisms that have been identified are flow boiling and condensation. A key strategy is to develop mechanistic models for flow conditions that would ensure insensitivity to gravity level for these important heat transfer mechanisms. This would allow performing most of the required confirmatory experiments on Earth. Those topics and the minimum flow requirements constitute key objectives for the proposed study. Unfortunately, most of the two-phase heat transfer know-how amassed over nearly a century of research comes from experiments conducted in Earth's gravity. The proposed study aims to develop an experimentally validated, mechanistic model for the minimum flow criteria to ensure gravity independent flow boiling heat transfer and condensation. Parabolic flight experiments and preparations for International Space Station (ISS) experiments will occur throughout the course of the proposed study. This project will be a joint effort between the Purdue University Boiling and Two-Phase Flow Laboratory (PU-BTPFL) and the NASA Glenn Research Center. Personnel from the two organizations combine extensive experience in research and development of both flow boiling and condensation systems, and in conducting microgravity experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed research will lay the foundation of development of advanced high power to weight ratio thermal management systems for NASA's future space vehicles and planetary bases.
	Part 1: Flow Condensation in Reduced Gravity. Vehicles for future manned space missions will demand unprecedented increases in power requirements and heat dissipation. Achieving these goals while maintaining acceptable size and weight limits will require replacing present single-phase thermal management components with far more efficient two-phase counterparts. This study discusses the development of an experimental facility for the study of annular condensation of FC-72 in microgravity, which was tested in parabolic flight as a prelude to the development of NASA's Flow Boiling and Condensation Experiment (FBCE) for the International Space Station (ISS). The flow behavior of the condensate film is shown to be sensitive mostly to the mass velocity of FC-72, with low mass velocities yielding laminar flow with a smooth interface, and high mass velocities turbulent flow with appreciable interfacial waviness. A select number of tests repeated in microgravity, Lunar gravity, and Martian gravity prove that the influence of gravity is very pronounced at low mass velocities, manifest by circumferential uniformity for microgravity versus appreciable thickening along one side of the condensation tube for Lunar and Martian conditions. However, the thickening is nonexistent for Lunar and Martian conditions at high mass velocities due to increased vapor shear on the film interface, proving high mass velocity is an effective means to negating the influence of gravity in space missions. For microgravity, the condensation heat transfer coefficient is highest near the inlet, where the film is both thin and laminar, and decreases along the condensation length, but increases again downstream for high mass velocities due to turbulence and increased waviness. A model is proposed to predict the condensation heat transfer which accounts for dampening of turbulent fluctuations near the film interface. The model shows good agreement with the heat transfer coefficient data in both trend and magnitude. Part 2: Experimental Measurement and Modeling of Downflow Condensation in a Circular Tube. Part 2 explores the interfacial and heat transfer characteristics of annular condensation of FC-72 in vertical downflow. Two separate condensation test modules are employed, one for high-speed video imaging of the film interface and the second for heat transfer measurements. Condensation in both test modules is achieved by rejecting the heat to a counterflow of cooling water. The heat transfer measurements are obtained along the inner wall of an 11.89-mm i.d. and 1,259.84-mm long stainless steel tube. For very low FC-72 flow rates, the film is observed to be both smooth and laminar. The film turns turbulent with a very wavy interface as the flow rate of FC-72 is increased, especially for exit film Reynolds numbers above 1,800. The heat transfer coefficient decreases axially because of a gradual thickening of the liquid film. However, the data show a downstream minimum before the heat transfer coefficient increases again towards the outlet as the film transitions to turbulent flow, enhanced by the more intense downstream waves. A control-volume-based model is proposed, which incorporates an eddy diffusivity profile for the liquid film that accounts for interfacial dampening of turbulence due to surface tension. The model shows good accuracy in predicting the average condensation heat transfer coefficient data, evidenced by a mean absolute error of 12.59%. Part 3: Climbing Film, Flooding, and Falling Film Behavior in Upflow Condensation in Tubes. Part 3 explores upflow condensation in vertical tubes. This orientation is complicated by the relative magnitude of the opposing vapor shear and gravity. This study examines the different flow regimes for condensation of FC-72 in a vertical tube using both high-speed video imaging and detailed heat transfer measurements. Four regimes are identified--falling film, where the condensing film drains downwards by gravity opposite to low velocity vapor flow, oscillating film, corresponding to film flow oscillating between upwards and downwards, flooding, where film begins to be sheared upwards by the vapor core, and climbing film, where high vapor velocity causes the film to be sheared upwards. The four flow regimes are well segregated in a flow regime map based on dimensionless superficial velocities of the vapor and liquid. The condensation heat transfer coefficient is shown to decrease axially because of gradual thickening of the film, except for high mass velocities, where turbulence and intensified interfacial waviness cause downstream heat transfer enhancement. An annular flow model is constructed, which shows fair predictions for the climbing film regime. The predictive accuracy of the model is influenced by flow oscillations occurring downstream of the climbing film region and inability of the model to account for interfacial waves. Part 4: Flow Boiling in Microgravity – A: Interfacial Behavior and Experimental Heat Transfer Results. Part 4 explores flow boiling of FC-72 in microgravity, which is simulated in a series of parabolic flight maneuvers. Flow boiling experiments are conducted in a rectangular channel fitted with two opposite heating walls. The operating conditions include liquid inlet velocities of 0.1 – 1.9 m/s, liquid mass velocities of 224.2 – 3347.5 kg/m2s, and inlet suboolings ranging from 2.8 – 8.1°C. The study includes both high-speed video analysis of interfacial features and heat transfer measurements. A dominant wavy vapor layer behavior is encountered for most operating conditions. Boiling is sustained mostly in 'wetting fronts' corresponding to contact regions between the wave troughs and the wall, and abated near the wave peaks. During a flight parabola, the heated wall temperatures decrease slightly as the aircraft enters the hypergravity ascent phase, then increase slightly during the microgravity phase, and decrease once again during the

Task Progress:	hypergravity descent. These temperature variations point to enhancement in flow boiling heat transfer with increasing gravity, and conversely a reduction with µge. Part 5: Flow Boiling in Microgravity – B: Critical Heat Flux Interfacial Behavior, Experimental Data, and Model. While Part 4 explored flow boiling conditions leading to critical heat flux (CHF), Part 5 addresses events just before CHF, during the CHF transient, and immediately following CHF. For both single-sided and double-sided heating, interfacial behavior just before CHF is characterized by dominant wavy vapor layers covering the heated walls, where liquid is able to access the walls only in the wetting fronts. CHF is associated with successive lift-off of wetting fronts from the walls, consistent with the Interfacial Lift-off Model, which has been validated extensively in past studies using single-sided heating in both µge and 1 ge. It is shown this model predicts µge double-sided flow boiling CHF with excellent accuracy. Additionally, the model points to convergence of CHF values for µge and 1 ge for inlet velocities greater than about 1 m/s. Therefore, maintaining velocities above this threshold allows designers of space systems to achieve inertia-dominated performance as well as to adopt prior data and correlations developed from terrestrial studies. Part 6: Consolidated Methodology to Predicting Flow Boiling Critical Heat Flux for Inclined Channels in Earth Gravity and for Microgravity. Part 6 addresses the development of a consolidated methodology to predicting flow boiling CHF for inclined channels in Earth gravity and for microgravity. It consists of a comprehensive investigation of the complex trends of flow boiling CHF in a rectangular channel in both gravitational environments. It is shown that the Interfacial Lift-off Model provides good predictions of CHF data for both single-sided and double-sided heating. CHF mechanism in Earth gravity is shown to be highly sensitive to flow orientation at very low velocities, but is consistent with the wavy vapor layer depiction of the Interfacial Lift-off Model at high velocities. The model predicts a stable vapor-liquid interface for downflow with a downward-facing heated wall at lower velocities, and wavy interface with a critical wavelength that decreases with increasing velocity at higher velocities. Predicted CHF values for microgravity fall about midway between the maxima and minima for Earth gravity. Overall, predicted values of CHF and key interfacial parameters for all orientations in Earth gravity and for microgravity converge around 1.5 m/s, which points to a velocity threshold above which inertia begins to effectively negate gravity effects. Part 7: Review of Flow Boiling Critical Heat Flux in Microgravity. Part 7 reviews published literature concerning two-phase flow and heat transfer in reduced gravity. Discussed are the different methods and platforms dedicated to exploring the influence of reduced gravity, including ground flow boiling experiments performed at different orientations relative to Earth gravity, as well as reduced gravity adiabatic two-phase flow, pool boiling, flow boiling and CHF experiments. Despite the extensive data and flow visualization results available in the literature, it is shown that there is a severe shortage of useful correlations, mechanistic models, and computational models, which compromises readiness to adopt flow boiling in future space systems. Key recommendations are provided concerning platform, heater design, and operating conditions for future studies to expedite the deployment of two-phase thermal management in future space missions. Part 8: Assessment of Body Force Effects in Flow Condensation - A: Experimental Investigation of Liquid Film Behavior for Different Orientations. Part 8 addresses the influence of flow orientation on condensation in tubes. Body force effects in flow condensation vary depending on channel orientation and fluid mass velocity, making the design of systems intended to operate in multiple orientations more complicated than those at a fixed orientation. This study examines the effects of body force on liquid film development for flow condensation of FC-72 in horizontal, vertical upflow, and vertical downflow orientations. Two test sections are utilized, one capable of providing high-speed imaging of liquid film development, and the other designed to allow detailed measurements of flow condensation heat transfer coefficient. High speed imaging shows that for low FC-72 mass velocities, flow regimes differ significantly among the three orientations, with vertical upflow exhibiting falling film behavior, horizontal flow showing stratification, and vertical downflow displaying annular co-current flow. For the case of low mass velocity horizontal flow, interfacial disturbances in the form of a sinusoidal wave are clearly visible with wavelengths on the order of 1-10 mm. As mass velocity is increased, the liquid film is seen to exhibit similar behavior for all three orientations due to interfacial shear stress negating body force effects. Heat transfer measurements reinforce these trends, with circumferential variations in heat transfer coefficient present for horizontal flow at low mass velocities, and differences in the axial variations in heat transfer coefficient seen when comparing vertical upflow to vertical downflow. As mass velocity is increased, differences in heat transfer coefficient are reduced, with the highest mass velocities exhibiting almost no variation with orientation. This convergence of values indicates the ability of interfacial shear stress to mitigate body force effects at sufficiently high mass velocities. Part 9: Assessment of Body Force Effects in Flow Condensation - B: Criteria for Negating Influence of Gravity. Part 9 concerns the development of a set of mechanistic criteria capable of predicting the flow conditions for which gravity independent flow condensation heat transfer can be achieved. Using FC-72 as working fluid, a control-volume based annular flow model is solved numerically to provide information regarding the magnitude of different forces acting on the liquid film and identify which forces are dominant for different flow conditions. Separating the influence of body force into two components, one parallel to flow direction and one perpendicular, conclusions drawn from the force term comparison are used to model limiting cases, which are interpreted as transition points for gravity independence. Experimental results for vertical upflow, vertical downflow, and horizontal flow condensation heat transfer coefficients are presented, and show that, for the given test section, mass velocities above 425 kg/m2s ensure gravity independent heat transfer. Parametric evaluation of the criteria using different assumed values of mass velocity, orientation, local acceleration, and exit quality show that the criteria obey physically verifiable trends in line with those exhibited by the experimental results. As an extension, the separated flow model is utilized to provide a more sophisticated approach to determining whether a given configuration will perform independent of gravity. Results from the model show good qualitative agreement with experimental results. Additionally, analysis of trends indicate use of the separated flow model captures physics missed by simpler approaches, demonstrating that use of the separated flow model with the gravity independence criteria constitute a powerful predictive tool for engineers concerned with ensuring gravity independent flow condensation heat transfer performance. EDITOR'S NOTE: Project continues with same Principal Investigator, Dr. Issam Mudawar, as "Flow Boiling and Condensation Experiment (FBCE) -- NNX17AK98G," from 6/1/2017-12/31/2021. See that project for subsequent reporting.
Bibliography Type:	Description: (Last Updated: 01/06/2026)

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